MTH 200 **Related Rates**

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. A ball of ice melts so that its surface area decreases at a rate of $2\,cm^2/\min$. Find the rate at which the diameter is decreasing when the diameter is $12cm$.

2. A rocket, rising vertically, is tracked by a radar station that is on the ground 5 miles from the launch pad. How fast is the rocket rising when it is 4 mile high and its distance from the radar station is increasing at a rate of 2000 miles/hour?

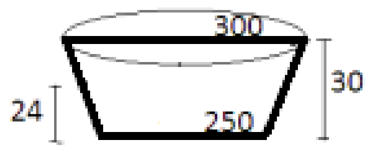
3. Wheat is poured through a chute at a rate of $10\ ft^3/\min$ and falls in a conical pile whose bottom radius is always half the altitude.

a) How fast will the radius of the base be increasing when the pile is 8ft high?

b) How fast will the circumference of the base be increasing when the pile is 8ft high?

c) How fast will the area of the base be increasing when the pile is 8ft high?

4. A 30ft deep retention pond is roughly shaped like a section of an inverted right circular cone, the bottom having a radius of 250ft and the top having a radius of 300ft (see the diagram below). During a heavy rainstorm, the pond has overcome its discharge tube and has risen to 24ft deep. At that instant the water level is still rising at the rate of about 2inches/hour. If the rain doesn't let up, when would you predict the pond to overflow its banks?



5. A camera is mounted at a point 100m away from a rocket launching pad. The rocket rises vertically at 28,000 km/hr.

a) The angle of elevation of the of the camera needs to change at just the right rate to keep it in sight. At what rate does the camera need to rotate upwards (in radians per second) when the rocket it 1 km off the ground?

b) In addition, to the angle of the camera needing to change, the camera-to-rocket distance is changing constantly, which means the focusing mechanism will also have to change at just the right rate to keep the picture sharp. A lens is able to focus according to the formula $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}$, where f represents the focal length of the camera lens, d_o represents the distance to the object we are trying to get in focus, and d_i represents the distance from the lens to the image (loosely speaking how far the lens has to move for the object to remain in focus).

i. At what rate is the distance between the rocket and camera increasing at the moment in time when the rocket is 1 km off the ground? Express your answer in m/s.

ii. Suppose that the camera lens being used has a focal length of 800mm. At what rate does the lens need to be moving/adjusting at the moment when the rocket is 1 km off the ground?